Higher-MI — Pruned — Full From Scratch — All Problems Fixed

A Unified Benchmark of Grokking Predictors — Predictor 7 of 8

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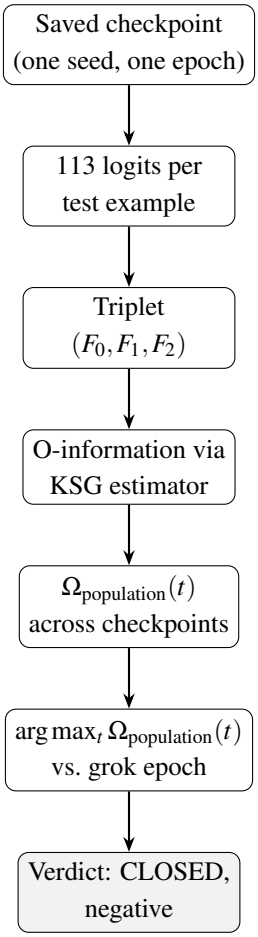
Roadmap of this document (12 chapters). Chapter 1 gives the whole pipeline in one diagram. Chapters 2–4 build the O-information from scratch on a simple, fully worked discrete example (the two-way switch). Chapter 5 defines F_0, F_1, F_2 from the raw task, one symbol at a time. Chapters 6–8 explain why F_0, F_1, F_2 need a continuous estimator (KSG) rather than the discrete counting of Chapters 2–4, and build that estimator. Chapter 9 documents two bugs found while implementing it. Chapter 10 fixes the event rule used to turn a curve into a single epoch. Chapter 11 gives the raw curves and the frozen-protocol numbers. Chapter 12 closes with what the negative result does and does not mean for the thesis.

Prerequisite. You have read the *Information_Theory_Intuitively* document (V31). This document assumes $H(X)$, $H(X \mid Y) = \sum_y p(y) H(X \mid Y = y)$, $I(X;Y) = H(X) - H(X \mid Y)$ and $S(x) = -\ln p(x)$ are already known, so none of these are re-derived here. This document starts directly from the point where conditioning can *increase* mutual information, and builds everything needed for Higher-MI — the O-information, the KSG estimator, the F0/F1/F2 triplet, the two implementation bugs, and the final frozen-protocol verdict — from that one starting point.

Consolidated predictor status (context, not re-derived here): L2 Norm, Dropout, Spectral, AGE, HTSR Alpha and Weight-PCA are all CLOSED, negative. Higher-MI (Predictor 7, this document) is also CLOSED, negative. Commutator Defect (Predictor 9) is next, not started.

Chapter 1 Big Picture: One Page Pipeline

Result: Higher-MI is CLOSED, negative. The picture below is the whole pipeline, checkpoint to verdict, in one pass. Every box in this diagram is built up from scratch in the chapters that follow.



So what: Each of the six boxes above is one chapter. Chapter 5 builds (F_0, F_1, F_2). Chapters 6–8 build the KSG-based O-information. Chapter 9 fixes two bugs found inside that pipeline. Chapter 10 defines the event rule. Chapter 11 gives the frozen-protocol numbers that produce the verdict. Chapter 12 states what the negative result means for the thesis.

Chapter 2 Synergy vs. Redundancy: Conceptual

Setup. Higher-MI asks a question about three quantities read off a trained network's output logits at a test example: the correct-class logit, the best wrong-class logit, and the average of the rest. The hypothesis is that how these three quantities share information with each other changes character across training — messy and *synergistic* during memorization, cleaner and *redundant* after generalization. Before touching any of that, this chapter fixes what synergy and redundancy mean, using two stories with no numbers yet. Chapter 3 returns to the second story and does the full numeric derivation.

Case (i): Redundancy — the fire alarm. A fire alarm panel reads two sensors: a smoke detector X and a heat detector Y. Both sensors are driven by one common cause Z = whether there is a fire. When there is a fire, both smoke and heat tend to rise together; when there is no fire, both tend to stay low together. Knowing X already tells you a great deal about Y, because both are echoes of the same underlying event Z. This is **redundancy**: X and Y overlap in what they tell you about each other, and this overlap does not go away once you already know Z — in fact a lot of what X and Y share, they share *because of* Z, so $I(X;Y)$ is comparatively large and conditioning on Z typically shrinks it further or leaves it similar. The O-information for this case comes out **positive**.

Case (ii): Synergy — the two-way staircase switch. A staircase light is wired with two independent switches: S_1 at the bottom, S_2 at the top. Each family member flips either switch independently of the other, so $X = S_1$ and $Y = S_2$ are statistically independent — knowing one switch's position tells you nothing about the other. The bulb state is $Z = X \oplus Y$ (XOR): the bulb is on exactly when the two switches disagree. Here is the surprising part. X and Y are independent, so $I(X;Y) = 0$ exactly. But the moment you are told Z — the bulb's state — X and Y become completely linked: if $Z = 1$ you know instantly that X and Y disagree, and if $Z = 0$ you know they agree. So $I(X;Y \mid Z)$ is large, even though $I(X;Y)$ alone was zero. Conditioning on Z *increased* the dependence between X and Y instead of decreasing it. This is **synergy**: the dependence is encoded jointly across X, Y, Z and cannot be seen by looking at any two of them alone. The O-information for this case comes out **negative**, and the full numeric value, derived from scratch in Chapter 3, is $\Omega = -0.6595$ nats.

So what: Redundancy: conditioning cannot increase, and usually decreases, shared information $\Rightarrow$ $\Omega > 0$. Synergy: conditioning increases shared information from a low or zero baseline $\Rightarrow$ $\Omega < 0$. The Higher-MI hypothesis for grokking is exactly this distinction applied to (F_0, F_1, F_2): early training behaves like the synergistic switch story, late training behaves like the redundant fire-alarm story, and the population O-information should show a maximum *before* the grokking transition if that story is right.

Chapter 3 O-Information: Definition and the Worked Synergy Example

Definition (O-information, three variables). For three random variables X, Y, Z, define

$$\Omega \;=\; I(X;Y) \;-\; I(X;Y \mid Z). \tag{A}$$

A positive Ω is read as net redundancy among the three variables; a negative Ω is read as net synergy.

Worked Example: the two-way switch, with numbers. Out of every 100 times a family member in the story flips the switches independently, record how often each switch ends up in each position. Use these recorded, non-trivial, distinct frequencies, which are given once and used throughout without rounding:

$$p(X=1)=\frac{43}{100}, \quad p(X=0)=\frac{57}{100}, \quad p(Y=1)=\frac{39}{100}, \quad p(Y=0)=\frac{61}{100}.$$

Term (i): $I(X;Y)$.

Why: Term (i) is the first piece of Main formula (A). It asks how much X and Y share *before* anyone is told the bulb state Z.

What: Because the two switches are flipped independently in the story, the joint frequency of any pair of positions is exactly the product of the two marginal frequencies.

Given that $Y=1$, the probability that $X=1$ will be:

$$p(X{=}1 \mid Y{=}1)=\frac{p(X{=}1,Y{=}1)}{p(Y{=}1)}=\frac{\frac{43}{100}\cdot\frac{39}{100}}{\frac{39}{100}}=\frac{43}{100}=p(X{=}1).$$

So what: The conditional equals the unconditional for every pair of values, so X and Y are independent exactly, not approximately. By the Result $I(X;Y)=0$ whenever X, Y are independent, Term (i) is

$$I(X;Y)=0 \text{ nats exactly.}$$

Sanity check: the same Term (i) result via entropy, $I(X;Y) = H(X) - H(X \mid Y)$.

Why: The prerequisite document defines $I(X;Y) = H(X) - H(X \mid Y)$. This sanity check confirms Term (i) a second, independent way, using that route instead of the conditional-probability route above.

Surprise table for X (one outcome per line):

x	$p(x)$	$S = -\ln p(x)$
1	$43/100$	$-\ln(43/100) = 0.8440$
0	$57/100$	$-\ln(57/100) = 0.5621$

$$H(X) = \frac{43}{100} \times 0.8440 + \frac{57}{100} \times 0.5621 = 0.3629 + 0.3204 = 0.6833 \text{ nats.}$$

Given that $Y = 1$, the probability that $X = 1$ will be: $p(X{=}1 \mid Y{=}1) = \dfrac{43/100 \times 39/100}{39/100} = \dfrac{43}{100}$, identical to the unconditional $p(X{=}1)$; the same holds for $Y = 0$. Since every conditional distribution of X given Y equals the marginal distribution of X, the conditional entropy equals the marginal entropy:

$$H(X \mid Y) = H(X) = 0.6833 \text{ nats.}$$

$$I(X;Y) = H(X) - H(X \mid Y) = 0.6833 - 0.6833 = 0 \text{ nats.}$$

So what: Both routes — the direct conditional-probability check and the entropy-difference formula — agree exactly: $I(X;Y) = 0$. This is the general Result "independent $\Rightarrow I(X;Y) = 0$" applied here only as a numeric check, not as the proof itself; the general proof (independent joint factorizes, so every conditional equals the marginal, so $H(X \mid Y) = H(X)$ for any independent pair) is exactly the prerequisite document's own Result, re-used here, not re-derived.

Term (ii): $I(X;Y \mid Z)$.

Why: Term (ii) is the second piece of Main formula (A). It asks how much X and Y share *once the bulb state Z is known*, averaged over the two bulb states.

What: Main formula (A) needs $I(X;Y \mid Z)$ expanded as a weighted average over $Z = 0$ and $Z = 1$:

$$I(X;Y \mid Z) = p(Z{=}0)\, I(X;Y \mid Z{=}0) \;+\; p(Z{=}1)\, I(X;Y \mid Z{=}1). \qquad (\mathrm{A}')$$

Sub-step: the four joint cells. Because X, Y are independent, every joint cell is the product of the two marginals. All four joints are distinct, as required, and are shown with their origin:

$$p(X{=}1,Y{=}1) = \frac{43}{100} \cdot \frac{39}{100} = \frac{1677}{10000}, \qquad p(X{=}1,Y{=}0) = \frac{43}{100} \cdot \frac{61}{100} = \frac{2623}{10000},$$

$$p(X{=}0,Y{=}1) = \frac{57}{100} \cdot \frac{39}{100} = \frac{2223}{10000}, \qquad p(X{=}0,Y{=}0) = \frac{57}{100} \cdot \frac{61}{100} = \frac{3477}{10000}.$$

So what: Out of 10000 trials, 1677 have both switches up, 2623 have only S_1 up, 2223 have only S_2 up, and 3477 have both switches down. These four numbers add to exactly 10000, which is the first check that the joint table is consistent.

Sub-step: the bulb-state marginals, origin shown.

What: $Z = 0$ (bulb off) happens exactly when the two switches agree, i.e. $(1,1)$ or $(0,0)$. $Z = 1$ (bulb on) happens exactly when they disagree, i.e. $(1,0)$ or $(0,1)$.

$$p(Z{=}0) = p(X{=}1,Y{=}1) + p(X{=}0,Y{=}0) = \frac{1677}{10000} + \frac{3477}{10000} = \frac{5154}{10000},$$

$$p(Z{=}1) = p(X{=}1,Y{=}0) + p(X{=}0,Y{=}1) = \frac{2623}{10000} + \frac{2223}{10000} = \frac{4846}{10000}.$$

So what: Out of 10000 trials the bulb is off 5154 times and on 4846 times. $5154 + 4846 = 10000$, second consistency check passed.

Sub-step (a): $I(X;Y \mid Z{=}0)$**.**
Why: Given $Z = 0$, the two switches are known to agree, so Y should be completely determined by X. This sub-step checks that numerically, through $H(X \mid Z{=}0)$ and $H(X \mid Y,Z{=}0)$, since $I(X;Y \mid Z{=}0) = H(X \mid Z{=}0) - H(X \mid Y,Z{=}0)$.
Given that $Z = 0$*, the probability that* $X = 1$ *will be (3-step, full origin):*

$$p(X{=}1 \mid Z{=}0) = \frac{\dfrac{1677}{10000}}{\dfrac{1677}{10000} + \dfrac{3477}{10000}} = \frac{\dfrac{1677}{10000}}{\dfrac{5154}{10000}} = \frac{1677}{5154}.$$

Given that $Z = 0$*, the probability that* $X = 0$ *will be:*

$$p(X{=}0 \mid Z{=}0) = \frac{\dfrac{3477}{10000}}{\dfrac{5154}{10000}} = \frac{3477}{5154}.$$

Surprise table for $X \mid Z{=}0$ (one outcome per line, fraction first inside the logarithm):

x	$p(x \mid Z{=}0)$	$S = -\ln p(x \mid Z{=}0)$
1	$1677/5154$	$-\ln(1677/5154) = 1.1228$
0	$3477/5154$	$-\ln(3477/5154) = 0.3936$

Result (entropy as average surprise):

$$H(X \mid Z{=}0) = \frac{1677}{5154}\left[-\ln \tfrac{1677}{5154}\right] + \frac{3477}{5154}\left[-\ln \tfrac{3477}{5154}\right]$$

$$= 0.3254 \times 1.1228 + 0.6746 \times 0.3936$$

$$= 0.3655 + 0.2655$$

$$= 0.6309 \text{ nats.}$$

Why: Now check $H(X \mid Y,Z{=}0)$: once $Z = 0$ is known *and* Y is known, is X still uncertain? *Given that* $Z = 0$ *and* $Y = 1$*, the probability that* $X = 1$ *will be:* $p(X{=}1 \mid Y{=}1,Z{=}0) = 1$, because $Z = 0$ forces agreement, so $Y = 1$ forces $X = 1$. Symmetrically $p(X{=}0 \mid Y{=}0,Z{=}0) = 1$. Every conditional probability here is exactly 1 or exactly 0, so its surprise is exactly 0 or undefined-times-zero (by the standard convention $0 \ln 0 = 0$).

$$H(X \mid Y,Z{=}0) = 0 \text{ nats exactly.}$$

So what: Once you know the bulb is off and you know one switch, the other switch carries no more surprise at all — it is forced.
Term (ii)(a):

$$I(X;Y \mid Z{=}0) = H(X \mid Z{=}0) - H(X \mid Y,Z{=}0) = 0.6309 - 0 = 0.6309 \text{ nats.}$$

Sub-step (b): $I(X;Y \mid Z{=}1)$.

Why: Given $Z = 1$, the two switches are known to disagree, so Y should again be completely determined by X (just the opposite value). This sub-step checks that the same way as sub-step (a).

Given that $Z = 1$, the probability that $X = 1$ will be (3-step, full origin):

$$p(X{=}1 \mid Z{=}1) = \frac{\frac{2623}{10000}}{\frac{2623}{10000} + \frac{2223}{10000}} = \frac{\frac{2623}{10000}}{\frac{4846}{10000}} = \frac{2623}{4846}.$$

Given that $Z = 1$, the probability that $X = 0$ will be:

$$p(X{=}0 \mid Z{=}1) = \frac{\frac{2223}{10000}}{\frac{4846}{10000}} = \frac{2223}{4846}.$$

Surprise table for $X \mid Z{=}1$ (one outcome per line):

x	$p(x \mid Z{=}1)$	$S = -\ln p(x \mid Z{=}1)$
1	$2623/4846$	$-\ln(2623/4846) = 0.6138$
0	$2223/4846$	$-\ln(2223/4846) = 0.7793$

$$\begin{aligned} H(X \mid Z{=}1) &= \frac{2623}{4846}\left[-\ln\tfrac{2623}{4846}\right] + \frac{2223}{4846}\left[-\ln\tfrac{2223}{4846}\right] \\ &= 0.5413 \times 0.6138 + 0.4587 \times 0.7793 \\ &= 0.3322 + 0.3575 \\ &= 0.6898 \text{ nats.} \end{aligned}$$

By the same forcing argument as sub-step (a) (disagreement plus one switch forces the other),

$$H(X \mid Y, Z{=}1) = 0 \text{ nats exactly.}$$

Term (ii)(b):

$$I(X;Y \mid Z{=}1) = 0.6898 - 0 = 0.6898 \text{ nats.}$$

Final substitution — Term (ii) into (A′):

$$\begin{aligned} I(X;Y \mid Z) &= p(Z{=}0)\, I(X;Y \mid Z{=}0) + p(Z{=}1)\, I(X;Y \mid Z{=}1) \\ &= \frac{5154}{10000} \times 0.6309 + \frac{4846}{10000} \times 0.6898 \\ &= 0.5154 \times 0.6309 + 0.4846 \times 0.6898 \\ &= 0.3252 + 0.3343 \\ &= 0.6595 \text{ nats.} \end{aligned}$$

Final substitution — Term (i) and Term (ii) into Main formula (A):

$$\Omega = I(X;Y) - I(X;Y \mid Z)$$

$$= 0 - 0.6595$$

$$= -0.6595 \text{ nats.}$$

So what: Ω is negative, confirming synergy: the two switches share nothing on their own ($I(X;Y) = 0$), but once the bulb state is known they become almost fully linked ($I(X;Y \mid Z) = 0.6595$). This is the exact numeric content behind the one-line claim made in Chapter 2.

Why the fire-alarm story would not have given this. In the fire-alarm case X, Y already share a large $I(X;Y) > 0$ before Z is known, because both are driven by the common cause Z. Conditioning on Z there removes shared information rather than creating it, so $I(X;Y \mid Z) < I(X;Y)$ and $\Omega > 0$ — the opposite sign, confirming that the fire-alarm story is redundant, not synergistic, and is the wrong story for what Higher-MI is looking for.

Chapter 4 Permutation Averaging

Why: Main formula (A), $\Omega = I(X;Y) - I(X;Y \mid Z)$, singles out Z as the "conditioning" variable. With three variables F_0, F_1, F_2 there is no reason to prefer conditioning on one of them over the other two — the choice of which variable plays the role of Z is arbitrary, and a different choice can give a slightly different numeric Ω due to estimation noise even though the three variables are symmetric in the underlying question of synergy versus redundancy.

What: There are $3! = 6$ orderings of three labels into the roles "first variable, second variable, conditioning variable". Writing $\Omega(X,Y \mid Z)$ for formula (A) with Z conditioned on, the six permutations are

$$\Omega(X,Y \mid Z),\ \Omega(Y,X \mid Z),\ \Omega(X,Z \mid Y),\ \Omega(Z,X \mid Y),\ \Omega(Y,Z \mid X),\ \Omega(Z,Y \mid X).$$

Definition (permutation-averaged O-information).

$$\Omega_{\text{avg}} = \frac{1}{6}\Big[\Omega(X,Y \mid Z) + \Omega(Y,X \mid Z) + \Omega(X,Z \mid Y) + \Omega(Z,X \mid Y) + \Omega(Y,Z \mid X) + \Omega(Z,Y \mid X)\Big].$$

So what: Averaging over all six role-assignments removes the arbitrary choice of which variable is treated as the conditioning variable, and gives one stable number per checkpoint instead of six slightly different ones. $\Omega(X,Y \mid Z)$ and $\Omega(Y,X \mid Z)$ are mathematically equal by the symmetry of $I(X;Y)$ in X,Y, so only three numerically distinct quantities are actually computed; each is used twice in the average. Higher-MI applies this averaging to (F_0, F_1, F_2) at every checkpoint, producing the single population-level value $\Omega_{\text{population}}(t)$ used from Chapter 6 onward.

Chapter 5 From One Checkpoint to Three Variables: the (F_0, F_1, F_2) Definition Chain

Why: Every later chapter uses F_0, F_1, F_2. Each must be defined once, in order, before it is used anywhere — no symbol appears here before its definition.

Term (i): Task.

What: The task is modular addition. $p = 113$ is prime. Input is a pair of residues (a, b) with $a, b \in \{0, 1, \ldots, 112\}$. Output is $c = (a + b) \bmod 113$.

Example: $(a, b) = (23, 45) \Rightarrow c = 68$, since $23 + 45 = 68 < 113$.

Term (ii): Test example and test set.

What: There are $113 \times 113 = 12769$ possible input pairs in total. 30% of these are held out as the test set: $\lfloor 0.3 \times 12769 \rfloor = 3830$ test examples. A test example is one of these 3830 pairs, not seen during training, used to check generalization.

Term (iii): Class.

What: The output c can be any of 113 values, 0 through 112. Each possible value is called a class. For a given test example with true answer $c = 68$, there is exactly one correct class (68) and 112 wrong classes.

Term (iv): Score and logit.

What: For one test example, the trained network outputs 113 real numbers, one per class, before the softmax is applied. The official name for this raw, pre-softmax number is the *logit*. A higher logit means the network believes that class more; a logit is not itself a probability.

Term (v): F_0, the correct-class logit.

Why: F_0 measures how strongly the network believes the one class that is actually correct.

What: For class c, F_0 is simply the logit the network assigns to class c.

$$F_0 = \text{logit}[c].$$

Given that $c = 68$, the value of F_0 will be: $F_0 = \text{logit}[68]$, e.g. $F_0 = 3.21$ for this particular test example.

So what: F_0 is the model's own confidence in the right answer, read directly off the raw score, before any normalization.

Term (vi): F_1, the runner-up logit.

Why: F_1 measures the strongest competing, wrong belief — the closest the network comes to being fooled.

What: F_1 is the maximum logit among the 112 wrong classes.

$$F_1 = \max_{j \neq c} \text{logit}[j].$$

Given that $c = 68$ is excluded, the value of F_1 will be: $F_1 = \max_{j \neq 68} \text{logit}[j]$, e.g. $F_1 = 2.05$, attained at class $j = 12$ for this test example.

So what: The gap $F_0 - F_1 = 3.21 - 2.05 = 1.16$ is the network's confidence margin over its best wrong guess on this one example.

Term (vii): F_2, the rest-mean logit.

Why: F_2 measures the general background level of the remaining wrong classes, once the correct class and the one strongest wrong class are both set aside.

What: After removing c and the runner-up class r, there remain $113-2=111$ classes. F_2 is their average logit.

$$F_2 = \frac{1}{111} \sum_{j \neq c,\, j \neq r} \text{logit}[j].$$

Given that $c = 68$ and $r = 12$ are both excluded, the value of F_2 will be: $F_2 = \frac{1}{111} \sum_{j \neq 68,12} \text{logit}[j]$, e.g. $F_2 = -0.34$.

So what: F_2 is the "noise floor" — what an average wrong class looks like once the one serious contender is already accounted for separately.

Main formula (A): from 113 **numbers to** 3 **numbers.**

What: Main formula (A) of this chapter collapses 113 per-example logits down to a triplet:

$$113 \text{ logits} \longrightarrow (F_0, F_1, F_2). \tag{A}$$

So what: One test example gives one triplet (F_0, F_1, F_2). The test set has 3830 test examples, so one checkpoint gives 3830 triplets. Treating F_0, F_1, F_2 as three random variables, each with 3830 sampled values across the test set, is exactly what lets Chapters 3–4's Ω machinery be applied to them — not to one example, but to the whole population of test examples at that checkpoint. This is where the name "$\Omega_{\text{population}}(t)$" comes from: it is the O-information of F_0, F_1, F_2 computed over the population of 3830 test examples, at training step t.

Chapter 6 The Continuous Problem

Why: Chapter 3's worked example used X, Y, Z that each take only two values, so probabilities could be counted directly out of 10000 trials. F_0, F_1, F_2 are not like that — each is a real-valued logit, and across 3830 test examples takes 3830 different real values, essentially never repeating exactly.

What: To reuse the entropy and mutual-information machinery of Chapters 2–4 directly, one would need $p(F_0)$, $p(F_0, F_1)$, and so on, which would require sorting the continuous logit values into bins and counting how many of the 3830 values fall into each bin.

So what: Binning has two problems here. First, the entropy of a genuinely continuous variable is not finite the way $H(X)$ was for a two-valued X — the differential entropy of a continuous distribution depends on the units of measurement and can be made arbitrarily large or small just by choosing a scale, so it is not directly comparable across checkpoints. Second, and more practically, the bin width is an arbitrary choice imposed by the person doing the counting, and different bin widths can silently give different $I(F_0; F_1)$ values from the exact same 3830 logits. Higher-MI needs a method that computes mutual information straight from the real-valued samples, with no bin width chosen by hand. That method is the KSG estimator, built up in the next two chapters.

Chapter 7 The KSG Estimator

Why: The Kraskov–Stögbauer–Grassberger (KSG) estimator computes mutual information between two continuous variables directly from a finite sample of points, using nearest-neighbour distances instead of a fixed bin width.

Term (i): the k-th neighbour distance.

What: For each sample point i among the $N = 3830$ joint points $(F_0^{(i)}, F_1^{(i)})$, find its k-th nearest neighbour in the joint (F_0, F_1)-space, using some distance metric. Call the distance to that k-th neighbour ε_i (half the side length of the smallest box, centred at point i, that contains exactly k other points).

Term (ii): marginal neighbour counts.

What: Still centred at point i, with the box side fixed at ε_i from Term (i), count how many of the other $N-1$ points fall within ε_i in the F_0 direction alone; call this count $n_x(i)$. Do the same in the F_1 direction alone to get $n_y(i)$.

Term (iii): the digamma function.

What: $\psi(k)$ denotes the digamma function, the logarithmic derivative of the Gamma function, $\psi(k) = \Gamma'(k)/\Gamma(k)$. It behaves like a smoothed version of $\ln(k-1)$ for integer k, and is the function that replaces $\ln(\cdot)$ throughout the KSG formula because it is the correct bias-corrected quantity for counting nearest neighbours rather than counting probability mass directly.

Main formula (A): KSG estimator for $I(F_0; F_1)$.

$$\widehat{I}(F_0; F_1) = \psi(k) - \frac{1}{N}\sum_{i=1}^{N}\Big[\psi\big(n_x(i)+1\big) + \psi\big(n_y(i)+1\big)\Big] + \psi(N).$$

So what: $\psi(k)$ and $\psi(N)$ are fixed constants once k and N are chosen. The only thing that varies from point to point is how crowded the two marginal directions are around that point, captured by $n_x(i)$ and $n_y(i)$. If F_0 and F_1 are independent, the joint k-th neighbour distance and the marginal neighbour counts tend to cancel so that $\widehat{I} \approx 0$; if they are strongly dependent, the marginal counts $n_x(i), n_y(i)$ tend to be smaller than what independence would predict, and $\widehat{I}$ comes out positive. This is computed directly on the 3830 real-valued logit pairs, with no binning anywhere.

Chapter 8 Conditional Mutual Information via KSG

Why: Main formula (A) of Chapter 3, $\Omega = I(X;Y) - I(X;Y \mid Z)$, needs $I(F_0;F_1 \mid F_2)$, not just $I(F_0;F_1)$. The two-variable KSG estimator of Chapter 7 must be extended to three variables.

Main formula (A): conditional-MI KSG estimator.

What: The conditional version replaces the two marginal spaces of Chapter 7 with four spaces: the full joint (F_0,F_1,F_2)-space used to find ε_i, the (F_0,F_2)-space, the (F_1,F_2)-space, and the F_2-space alone, each giving its own neighbour count around point i.

$$\widehat{I}(F_0;F_1 \mid F_2) = \psi(k) - \frac{1}{N}\sum_{i=1}^{N}\Big[\psi\big(n_{xz}(i)+1\big) + \psi\big(n_{yz}(i)+1\big) - \psi\big(n_z(i)+1\big)\Big].$$

So what: $n_{xz}(i)$ counts neighbours in the joint (F_0,F_2) space, $n_{yz}(i)$ in (F_1,F_2), and $n_z(i)$ in F_2 alone, all measured within the same ε_i fixed by the full three-variable joint space. Plugging this, together with the Chapter 7 estimator and its two other permutations, into Main formula (A) and then averaging over the six permutations of Chapter 4, gives $\Omega_{\text{population}}(t)$ for one checkpoint t.

Chapter 9 Two Bugs Found and Fixed

Why: Both bugs were found inside `higher_mi.py` and its wrapper in `run_nanda_benchmark.py`, while implementing the KSG machinery of Chapters 7–8 against the project's real saved checkpoints. Both are documented here because they would otherwise have silently produced wrong numbers.

Bug 1: strict versus inclusive neighbour counting.

What: The default nearest-neighbour search used an inclusive "$\leq$" comparison when counting how many points fall within radius ε_i in Term (ii) of Chapter 7. Because ε_i is itself defined as an exact distance to one of the sample points, an inclusive comparison re-counts that boundary point as being within the radius, which systematically inflates $n_x(i)$ and $n_y(i)$ and biases $\widehat{I}$ downward.

So what: The fix uses a radius infinitesimally smaller than ε_i (via `nextafter`), so the boundary point is correctly excluded, matching the strict-inequality convention the KSG estimator was derived under. As a side effect, switching to a tree-based nearest-neighbour search at the same time cut the computation from roughly 10.5 hours down to about 27 minutes per run.

Bug 2: silent test-set mismatch.

What: The benchmark's checkpoint-recompute wrapper rebuilds a test-set loader from the saved dataset whenever it needs one, rather than reusing the original training-time loader object. The Higher-MI wrapper built this loader without first calling `torch.manual_seed(seed)` and `np.random.seed(seed)`, while the original training run always seeds before building its loader. Because the train/test split itself is produced by a random shuffle, an unseeded rebuild of the loader silently produces a *different* 3830-example test set than the one the model actually generalized to during training.

So what: This does not crash — it just silently computes F_0, F_1, F_2 on the wrong 3830 examples, which is more dangerous than a crash because nothing flags it. The fix seeds explicitly with `torch.manual_seed(seed)` and `np.random.seed(seed)` before calling `get_dataloaders()` inside the Higher-MI wrapper, matching the precedent already used independently by the Spectral and AGE predictors' own train-loader rebuilds.

Chapter 10 Population vs. Temporal, and the Event Rule

Why: $\Omega_{\text{population}}(t)$ is one number per checkpoint, computed across all 3830 test examples at that checkpoint. A separate, exploratory quantity, Ω_{temporal}, is computed instead across a sliding window of checkpoints at one fixed test example (or small group), to check whether the population-level signal and a purely time-windowed signal are secretly the same thing in disguise.

Primary event rule.

What: The PRIMARY, scored event for Higher-MI is defined as the single checkpoint at which $\Omega_{\text{population}}(t)$ is largest over the whole run:

$$P_s = \arg\max_t \ \Omega_{\text{population}}(t).$$

So what: This is a *global maximum* rule, not a zero-crossing rule. An exploratory zero-crossing signal (the epoch where $\Omega_{\text{population}}(t)$ first crosses zero) was also logged, but it fires extremely early in every seed (epoch 200–1600), before the oscillatory structure of the curve has even developed, which marks it as an initialization artifact rather than a meaningful event — so it is excluded from scoring by design. A temporal-control maximum (the analogous argmax applied to Ω_{temporal}) is likewise logged only as an exploratory, non-scored control.

Grok epoch, and the contrast with Weight-PCA.

What: The grok epoch is defined project-wide as the first epoch at which test accuracy exceeds 0.9. This definition is shared by every predictor in the benchmark. What differs between predictors is only the rule used to pick the predictor's own event epoch: Weight-PCA uses a different event rule (its own frozen criterion from its own chapter), while Higher-MI's PRIMARY rule is specifically the global maximum of $\Omega_{\text{population}}(t)$ stated above. The two predictors are compared against the same grok-epoch definition, but are not expected to fire by the same mechanism.

Chapter 11 Raw Curves and the Frozen Verdict

Raw curves (5 seeds, 401 checkpoints each).
What: In every one of the 5 seeds, $\Omega_{\text{population}}(t)$ is highly oscillatory and quasi-periodic across the *entire* run, not only after grokking. It repeatedly spikes up from near zero and falls back, sometimes slightly negative, with the spike amplitude trending upward as training proceeds. There is no single, early, sharp rise that stands out from this oscillation before grokking in any seed.

Per-seed event vs. grok epoch.

Seed	Grok epoch	P_s (population Ω max epoch)	P_s/grok ratio	Before grok?
0	12987	15500	1.19	No
1	6921	8500	1.23	No
2	9510	20600	2.17	No
3	11019	12900	1.17	No
4	23747	32500	1.37	No

So what: In all five seeds the population-Ω maximum lands *after* the grok epoch, by a factor of $1.17\times$ to $2.17\times$ the grok epoch itself. Zero out of five seeds count as "before grok," even allowing the project's standard 100-epoch margin.

Frozen-protocol thresholds and results.
What: The project's frozen protocol scores three statistics against fixed thresholds, the same thresholds applied to every predictor in the benchmark:

Statistic	Threshold needed	Higher-MI's value	Pass/Fail
lead_frac	≥ 0.80	0.0000	Fail
ρ (Spearman)	≥ 0.90	0.7000	Fail
mean_rel_lead	$\in [0.05, 0.90]$	-0.4254	Fail

So what: lead_frac $= 0.0000$ means the PRIMARY event fires after grok, beyond the 100-epoch margin, in all 5 of 5 seeds — none count as early. $\rho = 0.7000$ means P_s does rank-correlate with the grok epoch across seeds (a higher grok epoch does tend to go with a higher P_s), but well below the 0.90 bar needed to call that correlation reliable enough to use as a predictor. mean_rel_lead $= -0.4254$ is large and negative, confirming on average that the event lands well after grok rather than before it.

Verdict: FAIL.
What: Higher-MI is CLOSED, negative — the 7th predictor (of the 8 remaining in the evaluation order, Correlation Traps having been removed earlier) to fail this benchmark's stricter, ex-ante early-warning criterion, after L2 Norm, Dropout, Spectral, AGE, HTSR Alpha and Weight-PCA.

Exploratory signals, informational only (not scored).

What: The zero-crossing epoch fires very early (200–1600) in all 5 seeds but with Spearman $\rho = -0.70$ against grok epoch — negatively correlated, an artifact of the curve starting near zero at initialization, not a meaningful early-warning candidate. The temporal-control maximum fires before grok in 4 of 5 seeds, but with $\rho = 0.00$ — no rank correlation with grok epoch at all across seeds. Neither exploratory signal is close to qualifying under the frozen protocol; they are reported for completeness, not as a second chance for a PASS.

Chapter 12 What the Negative Result Means

Why: A CLOSED-negative verdict is not the same as a wasted experiment. This closing chapter states, in one pass, what this particular negative result does and does not say about grokking.

What this result rules out.

What: Under this project's frozen protocol, grokking in $(a+b) \bmod 113$ is not preceded by a clean synergy-to-redundancy transition in the (F_0, F_1, F_2) logit triplet, as measured by O-information estimated via KSG. The raw curves additionally show the signal is persistently oscillatory across the whole run rather than concentrated around the grok transition, and its amplitude grows over time regardless of when grok actually occurs in a given seed — more consistent with the signal tracking general logit-margin growth than a grokking-specific phase change.

What this result does not rule out.

What: It does not say that information-sharing among output logits is irrelevant to grokking in general. It says this particular operationalization — this triplet, this O-information formula, this KSG estimator, this event rule — does not lead grokking under this project's thresholds. A different triplet of quantities, a different conditioning structure, or a differently defined event rule could in principle behave differently; this chapter's job is only to report what was actually tested, not to generalize beyond it.

Implementation lessons carried forward.

What: The two bugs fixed in Chapter 9 — inclusive-boundary neighbour counting, and an unseeded test-loader rebuild — are both the kind of bug that produces plausible-looking wrong numbers without crashing. Both fixes (strict-radius counting via `nextafter`, and explicit seeding before `get_dataloaders()`) are now the project's standing precedent for any future predictor that needs its own rebuilt loader or its own nearest-neighbour search.

Consolidated predictor status (8 predictors, this project's evaluation order).

#	Predictor	Status
1	L2 Norm	CLOSED, negative
2	Dropout	CLOSED, negative
3	Spectral	CLOSED, negative
4	AGE	CLOSED, negative
5	HTSR Alpha	CLOSED, negative
6	Weight-PCA	CLOSED, negative
7	**Higher-MI**	**CLOSED, negative (this document)**
8	Commutator Defect	NEXT, not started

So what: Correlation Traps was removed from this evaluation order earlier in the project and is not counted among the 8. Nothing about the Higher-MI result above changes the status of any other predictor's own code: Higher-MI's module (`higher_mi.py`) and the two bug fixes of Chapter 9 are additions, not changes, to the shared pipeline (`unified_measurements.py`, `run_nanda_benchmark.py`), and no other predictor's own file was touched while implementing or running Higher-MI.

One-line summary.

What: One number per checkpoint, $\Omega_{\text{population}}(t)$, built from a verified test set and a bias-corrected KSG estimator, peaks after grokking in 5 of 5 seeds, with lead_frac $= 0.00$, $\rho = 0.70$, mean_rel_lead $= -0.43$ — CLOSED, negative, consistent with every predictor closed before it in this benchmark.